

Solving the Zebra Logic Puzzle: A Constraint Satisfaction Problem Representation Analysis

Abstract

This report presents a formal analysis of the classical Zebra Puzzle by modeling it as a Constraint Satisfaction Problem (CSP). The primary objective is to evaluate how different representations affect the efficiency and clarity of the solution process. Two main formulations are compared: a house-centric representation and an attribute-based (positional) representation. The analysis shows that mapping attributes to integer positions enables more efficient constraint propagation and simpler formalization. The final solution identifies that the Norwegian drinks water in the first house, and the Japanese owns the zebra in the fifth house. More importantly, the study highlights that problem representation is a critical component of intelligent problem solving.

1. Introduction

1.1 Purpose

The purpose of this report is to model the Zebra Puzzle as a Constraint Satisfaction Problem (CSP) and to analyze different representations of the problem. The emphasis is not only on obtaining the correct solution but on understanding how representation influences computational efficiency and reasoning capability in artificial intelligence systems [1].

1.2 Key Question and Core Issue

The central question guiding this analysis is:

How does the choice of variables and domains in a CSP affect the ability of an algorithm to efficiently enforce constraints and find a valid solution?

This reflects a deeper issue in AI: the effectiveness of a solution depends not only on the algorithm but also on how the problem is formulated [2].

2. Problem Statement (Information)

The Zebra Puzzle consists of five houses arranged linearly, each with unique attributes: Color, Nationality, Candy preference, Drink and Pet.

Each attribute category contains five distinct values, and no value is repeated.

The system is constrained by a set of logical statements, such as:

- The Englishman lives in the red house
- The green house is immediately to the right of the ivory house

- The Norwegian lives in the first house
- Milk is drunk in the middle house

The objective is to determine: Where the zebra lives, and Where water is drunk

Formally, this is a CSP defined by: A set of variables (X), Domains (D) and Constraints (C).

3. CSP Representations and Comparative Analysis

To solve the Zebra Puzzle as a Constraint Satisfaction Problem (CSP), the problem must first be abstracted into variables, domains, and constraints. Different modeling choices lead to different computational behaviors. In this section, two main representations are presented and analyzed, followed by a justification of the preferred formulation.

3.1 Representation A: Attributes as Variables (Preferred CSP Formulation)

In this representation, each attribute (nationality, house color, drink, cigarette brand, and pet) is modeled as an independent variable. The domain of each variable corresponds to the house positions {1, 2, 3, 4, 5}.

-Variables and Domains

- Variables: Englishman, Spaniard, Norwegian, Ukrainian, Japanese, Red, Green, Ivory, Yellow, Blue, Water, Tea, Milk, OrangeJuice, Coffee, Hershey, KitKats, Smarties, Snickers, MilkyWays, Dog, Snails, Fox, Horse, Zebra
- Domain: $D = \{1, 2, 3, 4, 5\}$

-Constraints

Constraints are expressed as:

- All-different constraints within categories (e.g., all nationalities must occupy distinct houses)
- Equality constraints (e.g., Englishman = Red)
- Positional constraints (e.g., Green = Ivory + 1)
- Adjacency constraints (e.g., $| \text{Norwegian} - \text{Blue} | = 1$)

This representation benefits from a direct mapping between natural language clues and arithmetic constraints over integer domains.

3.2 Representation B: House-Centered Representation (Alternative Formulation)

In this formulation, houses are treated as the primary variables, and each house contains a full tuple of attributes.

-Variables

- House1, House2, House3, House4, House5

-Each house variable contains:

- Color
- Nationality
- Drink
- Pet
- Candy

-Domains

Each attribute takes categorical values, for example:

- $\text{Color} \in \{\text{Red, Green, Ivory, Yellow, Blue}\}$
- $\text{Nationality} \in \{\text{Englishman, Spaniard, Norwegian, Ukrainian, Japanese}\}$

-Constraints

Constraints are expressed as conditional statements across house indices, for example:

- If House_i contains Englishman, then House_i must contain Red
- If Green is in House_i, Ivory must be in House_{i-1}
- Adjacency requires pairwise comparisons across all house indices

This formulation results in more complex logical expressions and less direct constraint propagation.

3.3 Comparison of Representations

Aspect	Representation A (Attributes → Houses)	Representation B (Houses → Attributes)
Constraint expression	Simple arithmetic relations	Nested conditional logic
Domain structure	Numeric (1–5)	Categorical
Propagation efficiency	High	Lower
Compatibility with heuristics	Direct (MRV, AC-3)	Indirect
Clarity	High	Moderate

Representation A allows constraints to be expressed as numerical relations, which simplifies propagation and reduces ambiguity. Representation B, while intuitive in a descriptive sense, introduces unnecessary complexity in constraint evaluation.

3.4 Justification of the Preferred CSP Representation

The attribute-based representation (Representation A) is preferred because it provides a more structured and computationally efficient formulation of the problem. By mapping attributes to integer domains representing house positions, spatial relationships can be encoded using simple arithmetic constraints. This leads to a compact and uniform constraint system that is well-suited for standard CSP solving techniques.

In contrast, house-centered representations require conditional reasoning across multiple variables, which increases the complexity of constraint evaluation and reduces the effectiveness of constraint propagation techniques.

Furthermore, Representation A enables the direct application of classical CSP methods such as node consistency, forward checking, and heuristic variable selection strategies like Minimum Remaining Values (MRV). These methods rely on well-defined, finite, and structured domains, which are naturally provided by the numeric formulation.

Finally, the quality of the representation has a direct impact on the efficiency of the search process. A better abstraction reduces the effective branching factor and improves the ability of the solver to prune inconsistent states early. Therefore, Representation A is not only simpler conceptually but also significantly more efficient computationally, making it the most appropriate choice for solving this problem within a CSP framework.

4. Results

Solving the CSP using the preferred representation yields the following configuration:

House	Color	Nationality	Drink	Candy	Pet
1	Yellow	Norwegian	Water	Kit Kats	Fox
2	Blue	Ukrainian	Tea	Hershey	Horse
3	Red	Englishman	Milk	Smarties	Snails
4	Ivory	Spaniard	Orange Juice	Snickers	Dog
5	Green	Japanese	Coffee	Milky Ways	Zebra

Final Answers

- The zebra lives in **House 5 (Japanese)**
- Water is drunk in **House 1 (Norwegian)**

5. Implications and Consequences

The most significant implication of this analysis is that problem representation directly determines tractability.

A poor representation:

- Increases the search space
- Reduces pruning efficiency
- Leads to slower or infeasible computation

A well-designed representation:

- Enables efficient constraint propagation
- Reduces domain sizes early
- Improves scalability to larger problems

In real-world applications such as scheduling, logistics, or resource allocation, choosing an inadequate representation can make a solvable problem computationally intractable.

6. Conclusion

The Zebra Puzzle demonstrates a fundamental principle in artificial intelligence:

The intelligence of a system is not only in the algorithm, but in how the problem is represented.

While multiple CSP formulations are possible, the attribute-based representation is superior because it transforms logical constraints into simple arithmetic relations. This enables efficient reasoning and highlights the importance of abstraction in intelligent systems.

7. References

- [1] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Pearson, 2021.
- [2] N. G. Santiago and M. A. Jiménez, *Writing Formal Reports: An Approach for Engineering Students in 21st Century*, 3rd ed. Mayagüez, PR: University of Puerto Rico, Mayagüez Campus, 2002.

[3] "Zebra Puzzle," *Wikipedia*. [Online]. Available:
https://en.wikipedia.org/wiki/Zebra_Puzzle

8. AI Usage Disclosure

AI tools were used to support brainstorming and structural organization of the report. All reasoning, representations, and conclusions were independently verified to ensure technical correctness and compliance with academic standards.